



MATHEMATICS

TREND ANALYSIS

(2016-2014)

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Number System	11	10	12	13
2	HCF and LCM of Numbers	4	3	4	4
3	Decimal Fractions	-	1	1	-
4	Square Roots and Cube Roots	2	1	2	1
5	Time and Distance	7	10	7	5
6	Time and Work	1	6	1	4
7	Percentage	2	4	2	2
8	Simple Interest	1	-	2	1
9	Compound Interest	-	1	2	1
10	Profit and Loss	1	1	-	1
11	Ratio and Proportion	2	3	2	1
12	Logarithm	1	-	-	-
13	Basic Operations and Factorisation	11	5	5	1
14	HCF and LCM of Polynomials	4	3	4	4
15	Rational Expressions	-	-	-	1
16	Linear Equations	1	1	1	1
17	Quadratic Equations and Inequations	7	4	1	9
18	Set Theory	-	1	1	-
19	Measurement of Angles and Trigonometric Ratio	8	11	12	11
20	Height and Distances	2	2	4	1
21	Lines Angles	3	-	1	-
22	Triangles	12	2	3	7
23	Quadrilateral and Polygon	3	5	4	4
24	Circle	7	4	5	6
25	Locus	-	-	-	-
26	Area and Perimeter of Plane Figures	4	6	11	4
27	Surface Area and Volume of Solids	-	11	8	11
28	Statistics	6	5	5	7
Total		100	100	100	100

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NUMBER SYSTEM

Generally (10-12) questions have been asked from this chapter. Questions, from this section usually test your basic knowledge of numbers and are mostly based on various properties of multiplication and division. A good number of statement based questions have been asked from this chapter.



NUMBER SYSTEM

Numbers are collection of certain symbols or figures called digits. The common number system in use is decimal number system. In this system, we use ten symbols each representing a digit. These are 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. A combination of these figures representing a number is called a numeral.

Types of Numbers

1. **Natural numbers** Numbers which are used for counting i.e. 1, 2, 3, 4, ... are called natural numbers. The set of natural numbers is denoted by ' N '. Smallest natural number is 1 but we cannot find the largest natural number as successor of every natural number is again a natural number.
2. **Whole numbers** Natural numbers including zero are known as whole numbers. The set of whole numbers is denoted by W .
 - Every natural number is a whole number.
 - Zero (0) is the only whole number which is not a natural number.
3. **Even numbers** The numbers which are divisible by 2 are called as even numbers. e.g 2, 4, 6, 8, 10, In general these are represented by $2m$, where $m \in N$.
4. **Odd numbers** The number which are not divisible by 2 are called as odd numbers. e.g. 1, 3, 5, 7, 9, In general, these are represented by $(2m - 1)$, where $m \in N$.
5. **Prime numbers** Those numbers which are divisible by 1 and the number itself are known as prime numbers. e.g. 2, 3, 5, 7, ..., etc. are prime numbers.
 - If a number is not divisible by any of the prime numbers upto square root of that number, then it is a prime number.
 - 2 is the only even number which is prime.
 - The prime numbers upto 100 are : 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89 and 97 i.e. there are 25 prime numbers upto 100.

6. **Coprime numbers** Two natural numbers x and y are said to be coprime, if they do not have any common divisor other than 1.

e.g. (9, 2), (5, 6), (11, 15) are the pairs of coprime numbers.

- If x and y are any two coprimes, a number p is divisible by x as well as by y , then the number is also divisible by xy .
- Coprime are also called as relatively prime numbers.
- **Twin primes** Twin primes are pair of primes which differ by 2. e.g. (3, 5), (7, 9), (11, 13) etc.

7. **Composite numbers** A composite number is any number greater than one that is not a prime number. e.g. 4, 6, 8, 9, ... all are composite numbers.

- '1' is neither prime nor composite.

INTEGERS

The collection of positive numbers, negative numbers and zero are called integers. The set of integers is denoted by Z or I .

Thus, Z or $I = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$ is the set of integers. Every natural number and whole number is a part of integer. So, $N \subset W \subset I$.

Types of Integers

Integers are of three types

- Positive Integers** It is a set of all positive numbers. It is denoted by I^+ , $I^+ = \{1, 2, 3, 4, \dots\}$
- Negative Integers** It is a set of all negative numbers. It is denoted by I^- , $I^- = \{\dots, -3, -2, -1\}$
- Non-negative Integers** An integer that is either 0 or positive is called non-negative integer. $\{0, 1, 2, 3, \dots\}$

Note '0' is neither positive nor negative.

EXAMPLE 1. The smallest 3 digit prime number is

- a. 101 b. 103 c. 109 d. 113

Sol. a. The smallest 3 digit number is 100, which is divisible by 2.

$\therefore$ 100 is not a prime numbers.

$\sqrt{101} < 11$ and 101 is not divisible by 2, 3, 5 and 7.

$\therefore$ 101 is a prime number.

Hence, 101 is the smallest 3 digit prime number.

RATIONAL NUMBERS

The numbers which are expressed in the form of p/q where p and q are integers and are coprimes, $q \neq 0$ are called rational numbers.

The set of rational numbers is represented by 'Q'.

e.g. $\frac{3}{5}, \frac{-3}{7}, 7, -6$ are rational numbers.

- The decimal expansion of every rational number is either terminating or non-terminating repeating.

e.g. $\frac{1}{5} = 0.2$, $\frac{1}{3} = 0.333\dots$, $\frac{8}{44} = 0.181818\dots$ etc.

- The recurring decimal have been given a short notation as $0.3333 = 0.\bar{3}$, $0.181818 = 0.\bar{18}$

Note • Zero is a rational number, since we can write $0 = 0/1$.
• Every natural number, whole number and integer is a rational number.

IRRATIONAL NUMBERS

The number which cannot be expressed in the form p/q , where p and q both are integers and $q \neq 0$ are known as irrational numbers. The irrational numbers when expressed in decimal form are in non-terminating and non-repeating form. e.g. $\sqrt{2}, \sqrt{5}, \sqrt{7}$, 0.101005001, etc.

- Here, it is notable that exact value of π is not $\frac{22}{7}$ or 3.14, as $\frac{22}{7}$ is a rational number while π is irrational.

IMPORTANT FACTS

1. If $a + \sqrt{b} = x + \sqrt{y}$, where a and x are rational and $\sqrt{b}$ and $\sqrt{y}$ are irrational, then $a = x$ and $b = y$.
2. The sum or difference of a rational and an irrational number is irrational number.
3. The product of rational and irrational number is also an irrational number.
4. If we add, subtract, multiply or divide two irrational numbers, we may get an irrational number or rational number.

EXAMPLE 2. The rational number lying between $\sqrt{2}$ and $\sqrt{3}$ is

- a. $\frac{49}{28}$ b. $\frac{56}{35}$
c. $\frac{63}{45}$ d. $\frac{85}{68}$

Sol. b. We have, $\sqrt{2} = 1.414\dots$ and $\sqrt{3} = 1.732\dots$

$$\frac{49}{28} = \frac{7}{4} = 1.75, \quad \frac{63}{45} = \frac{7}{5} = 1.4$$

$$\frac{56}{35} = \frac{8}{5} = 1.6, \quad \frac{85}{68} = \frac{5}{4} = 1.25$$

Clearly, 1.6 lies between $\sqrt{2}$ and $\sqrt{3}$.

Hence, $\frac{56}{35}$ lies between $\sqrt{2}$ and $\sqrt{3}$.

REAL NUMBERS

The collection of all rational and all irrational numbers together forms the set of real numbers and is denoted by 'R'. Thus, all natural numbers, whole numbers, integers, rational and irrational numbers are real numbers.

Properties of Real Numbers

Properties of real numbers are as follows

General Properties of R

1. If x and y are two real numbers, then either $x > y$, $y > x$ or $x = y$.
2. If x and y are two real numbers, then
 - (i) $x > y \Rightarrow \frac{1}{x} < \frac{1}{y}$
 - (ii) $x > y \Rightarrow -x < -y$
 - (iii) $x > y \Rightarrow x + a > y + a$
 - (iv) $x > y \Rightarrow xa > ya$, when $a > 0$
3. If $xy = 0 \Rightarrow x = 0$ or $y = 0$

Properties of operations on R

Let '*' be any operation defined on R.

1. **Closure Property** If $a \in R$ and $b \in R$, then $a * b \in R$.
 2. **Associative Property** If $a, b, c \in R$, then $a * (b * c) = (a * b) * c$.
 3. **Commutative Property** If $a, b \in R$, then $a * b = b * a$.
 4. **Identity** Let 'I' be the identity, then $I * a = a * I = a \forall a \in R$.
 5. **Inverse** Let a' be the inverse of a , then $a' * a = a * a' = I \forall a \in R$.
- The above properties hold for addition and multiplication on R.
 - The additive identity of R is 0 and additive inverse of $a \in R$ is $-a$.
 - The multiplicative identity of R is 1 and multiplicative inverse of $a \in R$ is $1/a$.
 - R is closed for subtraction, if $a, b \in R$, then $a - b \in R$.
 - If $a, b, c \in R$, then $a \times (b + c) = (a \times b) + (a \times c)$ is the distributive property on R.

Absolute Value of a Real Number

The absolute value of a real number x is denoted by $|x|$. Thus, $|3| = 3$ and $|-4| = 4$.

$$\text{If } x \text{ is any real number, then } |x| = \begin{cases} x, & \text{when } x > 0 \\ 0, & \text{when } x = 0 \\ -x, & \text{when } x < 0 \end{cases}$$

e.g.

1. If $x = 5$, $|5| = 5$, if $x = 0$, $|0| = 0$
If $x = -5$, $|-5| = -(-5) = 5$

2. $|x - 2| = x - 2$, if $x > 2$, $|x - 2| = 0$, if $x = 2$
 $|x - 2| = 2 - x$, if $x < 2$

Some Properties of Absolute Values

1. $|x| \geq 0$ for all real x .
2. $|x| = a$ means $x = a$ or $x = -a$
3. $|x| > a$ means $x > a$ or $x < -a$
4. If $n = x^2$; $\sqrt{n} = \sqrt{x^2} = |x| = x$, if $x > 0 = -x$, if $x < 0$

EXAMPLE 3. Find the value of x which satisfy the inequalities $|x| \geq x$ and $2x - 1 > 3$.

- a. All positive number
- b. All positive number greater than 2
- c. All negative number less than -2
- d. All negative number

Sol. b. $|x| \geq x$ is true for all real values of x . Now, consider $2x - 1 > 3$ or $2x > 4$ or $x > 2$.
So, the solution set is all positive number greater than 2.

FACTORS IN SET OF INTEGERS

Let $a, b \in I$, we say that a is a factor of b , if there exists an integer p such that $b = ap$; in short we write a / b read it ' a ' divides ' b '.

If a is factor of b , then b is called a multiple of a .

Properties of factor and multiples

For real numbers a and b , if $b = ac$ then ' a ' is a factor ' b ' and we write a / b .

- (i) $a / b, b / c \Rightarrow a / c$ (This law is known as transitivity)
- (ii) $a / a, \forall a \in R$ (This law is known as reflexivity)
- (iii) a / b and $a / c \Rightarrow a / b + c$ and $a / b - c$
- (iv) If p is prime number and p divides ab where a, b are integers, then p divides ' a ' or p divides ' b '. Thus $p / ab \Rightarrow p / a$ or p / b .

EXAMPLE 4. How many factors of $2^5 \times 3^6$ are perfect squares?

- a. 9
- b. 12
- c. 18
- d. 4

Sol. b. Any factor of this number should be of the form $2^a \times 3^b$.

For the factor to be a perfect square a, b have to be even a can take values 0, 2, 4 and b can take values 0, 2, 4, 6.

$\therefore$ Total number of perfect squares = $3 \times 4 = 12$